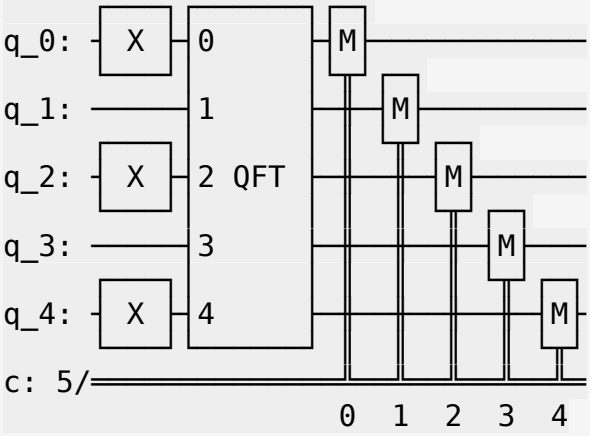
Practical 3

```
from qiskit import QuantumCircuit, transpile
from qiskit.circuit.library import QFT
from qiskit_aer import AerSimulator

qc = QuantumCircuit(5, 5)

# Encode sample state
qc.x(0)
qc.x(2)
qc.x(4)

# Apply QFT
qft = QFT(
    num_qubits=5,
    approximation_degree=0,
    do_swaps=True
)

qc.append(qft, range(5))

qc.measure(range(5), range(5))

simulator = AerSimulator()

compiled = transpile(qc, simulator)

result = simulator.run(compiled, shots=1000).result()

print("QFT Output:")
print(result.get_counts())

print("\nCircuit:")
print(qc.draw())
```

```
/tmp/ipykernel_3045/4268753117.py:13: DeprecationWarning: The class
``qiskit.circuit.library.basis_change.qft.QFT`` is deprecated as of
Qiskit 2.1. It will be removed in Qiskit 3.0. ('Use
qiskit.circuit.library.QFTGate or qiskit.synthesis.qft.synth_qft_full
instead, for access to all previous arguments.',)
  qft = QFT(

QFT Output:
{'10001': 32, '11101': 23, '11111': 29, '11011': 32, '10000': 33,
'01011': 28, '11110': 38, '11001': 31, '00000': 37, '00010': 31,
'00100': 27, '11100': 26, '00110': 28, '00001': 31, '10100': 31,
'10111': 34, '10010': 27, '10101': 29, '01111': 31, '11000': 45,
'00011': 32, '00111': 26, '01000': 33, '01100': 42, '10011': 25,
```

```
'01101': 30, '01010': 33, '01110': 35, '01001': 36, '00101': 31,
'11010': 27, '10110': 27}

Circuit:
```